DEPARTMENT OF INFORMATION TECHNOLOGY
COURSE: DATA COMMUNICATION

RESEARCH ON SATELLITE

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1. Introduction

Satellite communication is basically a way to send signals from one place on Earth to another through a satellite in space. It's like the satellite is a big mirror that catches the signal and bounces it back down to a different location, which can be really far away. This is a pretty powerful tool for talking to people over long distances. The idea is simple: a signal goes up to the satellite, and then it gets sent back down to where it's needed. This makes it possible to communicate with people who are thousands of kilometers away. It's a key part of how we stay connected today.

Satellite communication is really important because the Earth's surface is not flat and it's huge. There are lots of places that are hard to reach with regular phone or internet lines, like small islands with only a few hundred people, farms in the middle of big continents, science stations at the poles, and ships in the ocean. These places need to be able to communicate with the rest of the world, and satellites can help them do that. Satellites can reach areas that cables or towers can't, which is why they're so useful. As Roddy said in 2019, satellites are the best way to connect these remote areas to the rest of the world. They can provide phone and internet services to people who wouldn't be able to get them otherwise. This is especially important for people who live in really remote areas, because it helps them stay connected to the rest of the world and get the information and services they need.

It can be observed that satellite communication is not one uniform technology but a diverse family of systems, each designed for different purposes and operating at different altitudes. Some satellites sit so far from Earth that a single one can see almost half the planet at once. Others fly so close that they zip around the globe in 90 minutes, providing the low-latency internet connections that are changing how remote communities access digital services. Understanding the differences between these systems, and why those differences matter, is central to understanding satellite communication as a whole (Congressional Budget Office, 2023).

This essay aims to explore satellite communication in a thorough and accessible way. It covers the historical journey from early rockets to modern megaconstellations, explains the technical principles behind how satellites work, compares the different orbit types, examines frequency bands, discusses the advantages and disadvantages relative to terrestrial communication systems, addresses the major challenges the industry faces, and looks ahead at the trends that are likely to shape the next decade. Throughout, the goal is to present these ideas clearly and in plain language, supported by credible sources.

Satellites play a huge role in our daily lives, even if we don't realize it. For instance, the weather forecast you check on your phone is made possible by satellites that gather data from all around the world. Similarly, navigation apps that help you find your way to a new place rely on a network of satellites to provide location information. You can also watch live events happening on the other side of the world, thanks to satellites that transmit signals back to Earth. Moreover, satellites help provide internet access to rural areas, like schools, and enable emergency services to respond to distress calls from ships in trouble at sea. It's really worth learning more about how satellites work and how they impact our lives, as noted by NASA in 2023. By understanding satellite technology, we can appreciate the importance of these orbiting devices and how they make our lives easier and more connected.

2. The Evolution of Satellite Communications

2.1 The Road to Space: Early History

The history of satellite communication stretches back further than most people realise. Long before the first artificial satellite was launched, human beings were already working toward the technologies that would eventually make it possible. The Chinese invention of gunpowder around 300 BC and its eventual application to early rocket-like devices in the thirteenth century AD is considered one of the earliest chapters in that story. By the battle of Kai-Keng in 1232, Chinese defenders were already attaching primitive rockets to arrows, creating what were effectively the first guided projectiles (Encyclopedia.com, 2023).

Rocket science did not truly emerge as a serious discipline until the early twentieth century. American engineer Robert H. Goddard, working in the 1900s, received his first patents for liquid-fuelled rocket engines in 1914 and made the first successful launch of a liquid-fuelled rocket in 1926. Though his ideas were initially dismissed by the scientific establishment, his 200 patents eventually formed the foundation for the rocketry programmes of multiple nations. His 1929 launch, which carried a barometer and camera as scientific payloads, is considered the first data-gathering rocket flight in history (Encyclopedia.com, 2023).

It can be observed that the Second World War dramatically accelerated rocket development, as military powers on both sides recognised the strategic potential of long-range projectiles. The Cold War that followed between the United States and the Soviet Union transformed that military impetus into a race for space. Advances in solar cell technology, transistor miniaturisation, and digital computing all converged in the 1950s to make the launch of artificial satellites technically feasible for the first time (Pratt et al., 2019).

On October 4, 1957, the Soviet Union changed the world by launching Sputnik 1, a polished metal sphere roughly the size of a basketball, into low Earth orbit. It transmitted a radio beeping signal that could be picked up by amateur radio operators around the world, and it completed one orbit of the Earth every 96 minutes. Although Sputnik only functioned for 92 days before its batteries died and it eventually re-entered the atmosphere, its psychological and political impact was enormous. It demonstrated that space was not out of reach and triggered an urgent response from the United States (Encyclopedia.com, 2023).

The concept of geostationary orbit that would become central to the satellite communication industry had actually been described theoretically ten years earlier. In October 1945, British physicist and science fiction writer Arthur C. Clarke published an article in Wireless World magazine describing a system of satellites positioned 35,786 kilometres above the equator that would orbit the Earth in exactly 24 hours, appearing stationary from the ground. Clarke argued that just three such satellites, placed at equal intervals around the equator, could provide communication coverage to the entire surface of the Earth except for the polar regions. This vision, which Clarke called the Clarke Orbit or geostationary orbit, became the blueprint that the satellite industry would follow for the next five decades (Maral et al., 2020).

2.2 Commercial Development and the Rise of Broadcasting

The United States responded to Sputnik with the creation of NASA in 1958 and a series of increasingly sophisticated satellite launches. Early American communication satellites included SCORE, which transmitted a recorded Christmas message from President Eisenhower in 1958, and Courier, launched in 1960, which became the world's first fully active communications satellite, capable of receiving a signal from the ground and retransmitting it in real time. Neither lasted long, but they proved the concept (Encyclopedia.com, 2023).

The commercial era of satellite communication began in earnest in 1962 when NASA launched AT&T's Telstar 1, the world's first privately owned communications satellite. On July 10, 1962, Telstar transmitted the first live transatlantic television images, sending footage between the United States and receiving stations in the United Kingdom and France. Voice calls, data, and facsimile transmissions were also demonstrated. This single event opened space to commercial users and sparked what can fairly be described as a communications revolution (Encyclopedia.com, 2023).

From my understanding, the next milestone was the achievement of true geostationary orbit. In February 1963, Hughes Aircraft launched Syncom, the first satellite to reach geosynchronous altitude. After Syncom suffered an equipment failure, Syncom II was launched and then Syncom III, which in 1964 became the first satellite to transmit live television coverage of the Olympic Games from Tokyo to audiences in the United States. The ability to watch events from the other side of the world in real time was something that no previous technology had made possible (Pratt et al., 2019).

In April 1965, the first commercial communications satellite, Early Bird (also known as Intelsat I), was placed into geostationary orbit over the Atlantic Ocean. It provided the first continuous satellite communications link between North America and Europe and could carry 240 telephone conversations or one television channel simultaneously. Five years later, over half a billion people watched the Apollo 11 Moon landing live on television, a broadcast made possible by a network of Intelsat satellites. This moment demonstrated better than any technical specification what satellite communication could mean for humanity (Encyclopedia.com, 2023).

Throughout the 1970s and 1980s, domestic satellite systems multiplied rapidly. Canada launched Anik in 1972, becoming the first country to operate a domestic geostationary satellite. The United States followed with Western Union's Westar in 1974. Home Box Office began using satellites to distribute cable television content in 1975, triggering an explosion of cable channels and transforming entertainment. By the 1990s, direct broadcast satellite services were delivering hundreds of television channels directly to small dishes on the rooftops of millions of homes (Ilcev, 2017).

2.3 The LEO Revolution and the Modern Era

The late 1990s saw the first serious attempts to use large constellations of LEO satellites for mobile telephone services. Companies like Iridium and Globalstar raised billions of dollars to build these networks, promising truly global mobile phone coverage. The ambition was extraordinary, but the timing was unfortunate. The rapid growth of terrestrial cellular networks in the years between when these LEO systems were conceived and when they actually launched meant that the market had largely moved on by the time the satellites were operational. Both Iridium and Globalstar went bankrupt, though both were eventually restructured and continue to operate niche services today (Ilcev, 2017).

This suggests that the first LEO era failed not because the technology was wrong, but because the business model could not compete with the rapidly improving economics of ground-based mobile networks. The lesson was remembered, and the second LEO era, beginning in the 2010s, approached the market differently. Rather than targeting satellite phones as a replacement for cellular networks, companies like SpaceX focused on broadband internet as a use case where satellites genuinely had an advantage over terrestrial alternatives, specifically in rural and remote areas where no competitive broadband existed (Congressional Budget Office, 2023).

By 2024, the satellite industry had been fundamentally transformed. There were approximately 10,000 active satellites in orbit, compared to around 1,000 just ten years earlier. SpaceX's Starlink constellation alone accounted for roughly 60 percent of all active satellites. The satellite broadband market, once considered a niche service for remote oil platforms and maritime vessels, was now serving residential customers in rural areas of dozens of countries, competing directly with traditional internet service providers in speed and reliability if not yet in price (Goldman Sachs, 2024).

3. Technical Principles of Satellite Communication

3.1 How a Satellite Link Works

So, basically, when we talk about satellite communication, there are three main parts that work together. First, you have a ground station that sends a signal up to space. Then, there's the satellite itself, which gets the signal, makes it stronger, and sends it back down. Finally, you have another ground station or a user terminal that catches the signal coming back down. The part where the signal goes up is called the uplink, and the part where it comes back down is called the downlink. When we're talking about two-way communication, like when you're having a conversation, each user terminal is actually doing both jobs at the same time - it's sending and receiving signals simultaneously.

The satellite has special devices called transponders that can receive and send radio signals. These transponders get a signal from the ground, change its frequency so it doesn't interfere with other signals, make it stronger, and then send it back to Earth. The satellite gets the power it needs to do this from big solar panels that turn sunlight into electricity. But when the Earth blocks the sunlight, the satellite uses batteries to keep working.

The way a satellite's antenna is set up is really important for how well it works. Old satellites used simple antennas that sent out signals in all directions, but newer ones use special antennas that focus their signals on specific areas on the ground. These areas are called footprints. Some satellites, like the high-throughput kind, take it a step further by using lots of narrow beams that cover small areas, which lets them reuse the same frequencies in different beams without interfering with each other. This makes the whole network work a lot better and can handle a lot more traffic.

Setting up and keeping a good connection with a satellite is a tough job that needs to solve many technical problems at the same time. The signal has to be strong enough to travel a very long distance without getting too weak. It also has to be pointed directly at the satellite, which means the antenna on the ground has to follow the satellite's movement if it's not staying still. The signal has to be sent in a way that allows any mistakes made during transmission to be found and fixed. And the whole system has to be able to adjust to changes in the atmosphere that can affect the signal quality at any time. This is a big challenge, as noted by Pratt and his team in 2019. They pointed out that all these things have to work together for the satellite link to be reliable. If one thing goes wrong, the whole connection can be lost. So, it's a complex problem that needs careful planning and execution to get it right.

3.2 The Three Main Orbit Types

From my understanding, the altitude at which a satellite orbits determines almost everything about its performance characteristics. The three main orbital categories used in communication satellites are Low Earth Orbit, Medium Earth Orbit, and Geostationary Earth Orbit. Each represents a different set of trade-offs between coverage area, latency, satellite size, constellation size, and cost (Scott, 2024).

Geostationary Earth Orbit satellites sit at an altitude of approximately 35,786 kilometres directly above the equator. At this specific altitude, the time it takes for a satellite to complete one orbit exactly matches the time it takes for the Earth to rotate once on its axis, which is approximately 24 hours. The result is that the satellite appears to hang motionless in the sky when viewed from the ground. This is enormously convenient for applications like television broadcasting and weather monitoring, because a receiving dish only needs to be pointed at the satellite once and never adjusted again. A single GEO satellite can cover an area spanning almost an entire hemisphere (Maral et al., 2020).

The disadvantage of GEO satellites is the latency their altitude creates. A radio signal travels at the speed of light, approximately 299,792 kilometres per second. Even at that speed, the round trip from the ground to a GEO satellite and back covers roughly 72,000 kilometres, which takes about 240 milliseconds one way and approximately 480 to 600 milliseconds for a complete round trip when processing delays are included. For a telephone call, this creates a noticeable echo effect. For interactive applications like video games or financial trading systems, it renders satellite connections essentially unusable (Pratt et al., 2019).

Low Earth Orbit satellites operate between approximately 300 and 2,000 kilometres above the Earth's surface. Their proximity dramatically reduces the time signals take to make the round trip. Starlink, for instance, reports typical latency of 20 to 50 milliseconds, which is comparable to many cable broadband connections. The trade-off is that LEO satellites move very fast relative to the ground, completing a full orbit in about 90 to 120 minutes. A user's terminal must continuously hand off from one satellite to the next as each one passes overhead. To ensure that at least one satellite is always visible from any given point on Earth, operators must deploy constellations of hundreds or thousands of satellites (Congressional Budget Office, 2023).

Medium Earth Orbit covers the large range between roughly 2,000 and 35,786 kilometres. The most important application of MEO satellites is the Global Positioning System, which uses a constellation of 24 operational satellites orbiting at approximately 20,200 kilometres. MEO satellites offer a middle ground between GEO and LEO: their latency is typically in the range of 30 to 120 milliseconds, which is better than GEO but not quite as good as the best LEO systems.

Their coverage area per satellite is larger than LEO, meaning fewer satellites are needed to cover the globe. SES's O3b constellation, operating at around 8,000 kilometres, was an early example of MEO being used for broadband communications, particularly for maritime and island communities (Pratt et al., 2019).

Table 1: Comparison of the Three Satellite Orbit Types

Characteristic	LEO	MEO	GEO
Altitude	300-2,000 km	2,000-35,786 km	~35,786 km
Latency	20-50 ms	30-120 ms	500-700 ms
Coverage per satellite	Small footprint	Medium footprint	Near-hemisphere
Satellites needed for global coverage	Hundreds to thousands	Dozens	3 (near-global)
Orbital period	~90-120 minutes	2-12 hours	~24 hours
Typical applications	Broadband internet, IoT, imagery	GPS, navigation, maritime broadband	TV broadcast, weather, telephony
Example systems	Starlink, OneWeb, Kuiper	GPS, O3b	Viasat, DirecTV, Intelsat

3.3 Satellite Frequency Bands

Satellite frequency bands can be understood as the specific slices of the radio spectrum that a satellite uses to carry its signals. Different bands behave differently in the atmosphere, carry different amounts of data, and require different antenna sizes. Choosing the right frequency band for a satellite application involves balancing several competing requirements, including throughput, reliability, regulatory constraints, and hardware cost (Kratos Defense, 2023).

The C-band, covering approximately 4 to 8 GHz, was the first frequency range widely adopted for commercial satellite communications. Its relatively low frequency means that C-band signals are resilient against rain fade, the signal degradation caused by heavy precipitation. This makes C-band a reliable choice for applications where consistent service quality is critical regardless of weather conditions. The trade-off is that C-band requires larger dish antennas, typically several metres in diameter, and the available bandwidth is more limited than at higher frequencies. Many legacy satellite television and telephony systems continue to use C-band because of its reliability and the large installed base of compatible equipment (Kratos Defense, 2023).

The Ku-band, spanning roughly 12 to 18 GHz, became the dominant choice for the satellite television industry and later for VSAT networks. Ku-band antennas can be significantly smaller than C-band antennas, with residential dishes typically around 45 to 90 centimetres in diameter. This smaller size and lower hardware cost made Ku-band the foundation of the direct broadcast satellite television industry, with services like DirecTV in the United States and Sky in Europe delivering hundreds of channels directly to consumer homes. Ku-band is more susceptible to rain fade than C-band, but careful link engineering can manage this for most locations (Pratt et al., 2019).

The Ka-band, covering approximately 26.5 to 40 GHz, represents the frontier of commercial satellite communications. The much higher frequencies available in this band mean that Ka-band systems can carry far more data per unit of spectrum than C or Ku band systems. Most modern high-throughput satellite systems and all of the major LEO broadband constellations operate primarily in the Ka-band. Starlink, for instance, uses Ka-band for its gateway links between ground stations and satellites, alongside Ku-band for user terminals. The limitation of Ka-band is its heightened sensitivity to atmospheric conditions. Rain, heavy cloud cover, and even humid air can attenuate Ka-band signals significantly, requiring careful fade mitigation strategies (Viasat, 2022).

This suggests that frequency selection is never a simple decision. Engineers must weigh the data capacity needs of the application, the expected weather conditions in the service area, the cost and size constraints on user equipment, the regulatory environment governing spectrum use in each country, and the competitive landscape of what spectrum is actually available for new operators to use. In many cases, modern satellite systems use multiple frequency bands simultaneously, routing different types of traffic over different frequencies depending on their requirements (ITU, 2023).

3.4 High-Throughput Satellites

High-throughput satellites represent a fundamental architectural shift in how satellites are designed to deliver capacity. In conventional satellite design, a single broad antenna beam covers a large geographic area, and all users within that area share the available bandwidth. The total capacity of such a satellite is limited by the spectrum it has been allocated and the power of its transponders. A high-throughput satellite instead divides its coverage area into dozens or even hundreds of narrow spot beams, each serving a much smaller geographic area (OmniAccess, 2020).

It can be observed that the key insight behind spot beam technology is frequency reuse. Because each spot beam covers only a small area, two spot beams that are far enough apart can use exactly the same frequencies without interfering with each other. This means the total capacity of the satellite is the product of its per-beam capacity multiplied by the number of non-interfering beams it can operate simultaneously. A satellite with 100 spot beams reusing frequencies can, in principle, deliver 50 or more times the capacity of a conventional single-beam satellite using the same spectrum allocation (Viasat, 2022).

ViaSat-1, launched in October 2011, made this tangible. At the time of its launch, it offered more than 140 Gbits per second of total network capacity, more than all other commercial communications satellites over North America combined. This achievement demonstrated that satellite broadband could deliver speeds and economics that were genuinely competitive with terrestrial broadband alternatives, at least in terms of capacity cost. The cost per bit transmitted via ViaSat-1 was estimated to be less than $3 million per Gbps, compared to over $100 million per Gbps for conventional Ku-band capacity at the time (Amos, 2011).

From my understanding, the high-throughput satellite concept has now been embraced across the industry and has extended beyond geostationary orbit. LEO satellites, despite their smaller size, can use the same spot beam technology and additionally benefit from their proximity to Earth to reduce the signal power required. The SES O3b constellation, operating in medium earth orbit, was the first MEO high-throughput satellite system, launching in 2013 and proving that the technology could work outside of geostationary orbit. Today, LEO megaconstellations like Starlink are essentially networks of many small high-throughput satellites working together (SatNews, 2018).

4. The New LEO Space Race

4.1 The Scale of the Current Build-Out

From my understanding, what is unfolding in the satellite industry today is without historical precedent in terms of its scale and pace. The transition from a world of a few hundred geostationary satellites to one of potentially tens of thousands of LEO satellites is happening within a single decade. Goldman Sachs Research has forecast that as many as 70,000 LEO satellites could be launched within five years, a number that would have seemed fantastical even ten years ago (Goldman Sachs, 2024).

SpaceX's Starlink constellation is the dominant player in this space, having launched more than 7,800 satellites by 2025 and operating plans for a final constellation of 42,000 satellites. The company has demonstrated a launch cadence that no previous operator has approached, using its own Falcon 9 rockets to deploy batches of satellites at a cost per kilogram to orbit that has fallen dramatically compared to historical norms. This cost reduction, made possible by rocket reusability, is one of the central reasons why LEO constellation economics have become viable when they were not in the 1990s (SpaceX, 2025).

It can be observed that competition is intensifying across multiple fronts. Amazon's Project Kuiper received FCC approval to launch a constellation of over 3,000 satellites, with its first commercial satellites reaching orbit in 2025. Eutelsat's OneWeb constellation, which emerged from bankruptcy and was later acquired with British government support, operates over 630 satellites. Telesat's Lightspeed project, backed by a substantial Canadian government loan, aims to build a high-performance LEO network optimised for enterprise and government customers. Iridium, the original LEO mobile satellite company from the 1990s, continues to operate its second-generation constellation of 80 satellites (Congressional Budget Office, 2023).

This suggests a market that is becoming significantly more competitive. Where satellite broadband was once dominated by a small number of GEO operators charging premium prices for limited capacity, the LEO era is bringing more providers, more capacity, and downward pressure on pricing. The implications for rural and remote communities, maritime and aviation customers, and government and military users are all significant (Congressional Budget Office, 2023).

4.2 National and Governmental Involvement

Lots of governments are now getting involved in the low Earth orbit satellite business, and it's not just about making sure everyone follows the rules. Many countries think that having their own space-based communication systems is crucial for national security and strategic interests. Because of this, governments are putting money into commercial satellite programs that might not have been possible otherwise, simply because they wouldn't have been profitable enough on their own. This is a big deal, and it's changing the way the satellite industry works - as noted by the Congressional Budget Office just this year.

The European Union launched its IRIS2 project in December 2024, awarding a major contract to build a multi-orbit constellation of 290 satellites designed to provide connectivity for government users, critical infrastructure, and eventually commercial broadband customers. The project is explicitly designed to reduce European dependence on non-European satellite operators for sensitive communications. In China, two megaconstellation projects, Guowang and Qianfan, have filed for permission to launch 13,000 and 14,000 satellites respectively, and China's government has made it clear that space-based communication is a strategic priority (Congressional Budget Office, 2023).

The rules set by the International Telecommunication Union, or ITU, about who gets to use which parts of the radio spectrum have become a big deal in international politics. You see, companies that want to launch satellites have to tell the ITU which parts of the spectrum they want to use, and they have to actually put up their satellites within a certain timeframe or they lose their claim. This has created a rush to not only launch satellites, but to also secure the right to use the spectrum they need to operate. Countries that are quick to file their claims and get their satellites up in space get a big advantage over those that are slower to act, and it's hard for latecomers to catch up.

5. Comparative Analysis: Satellite Versus Terrestrial Communication

5.1 Advantages of Satellite Communication

It can be observed that the most compelling arguments for satellite communication are not about competing with terrestrial networks in their strongest markets, but about serving needs that terrestrial networks cannot adequately address. The following advantages reflect where satellite communication genuinely excels (Maral et al., 2020).

- **Reach into remote and underserved areas:** Satellites can provide communication services to any location on Earth with a clear view of the sky. This includes remote farming communities, isolated islands, polar research stations, ships at sea, aircraft in flight, and areas of developing countries where terrestrial infrastructure has never been economically viable to build. For these locations, satellites are often the only realistic option for connectivity (Maral et al., 2020).
- **Resilience during natural disasters:** Ground-based infrastructure is vulnerable to floods, earthquakes, hurricanes, and wildfires. When these events strike, the telephone lines, cell towers, and fibre cables that communities depend on may be destroyed or rendered inaccessible. Satellite communication can remain operational even when everything on the ground has been damaged, making it a critical tool for emergency response and disaster recovery (NASA, 2023).
- **Independence from terrestrial infrastructure:** Because satellites operate in space, they are not affected by the condition of roads, bridges, or power grids on the ground. A satellite link can be established in a completely new location within hours, requiring nothing more than a terminal, a power source, and a clear view of the sky. This flexibility makes satellites invaluable for military operations, humanitarian missions, and any situation where rapid communications deployment is needed (Ilcev, 2017).
- **Simultaneous broadcasting to many receivers:** A single satellite can transmit the same signal to an unlimited number of receiving terminals simultaneously. This point-to-multipoint capability is uniquely suited to broadcasting applications like television, radio, and distribution of software or data updates to many devices at once. No terrestrial technology can match the efficiency of a satellite for this specific use case (Pratt et al., 2019).
- **Connectivity for mobile platforms:** Ships crossing oceans, aircraft flying long international routes, trains travelling through countryside without cellular coverage, and remotely operated vehicles in hostile environments can all maintain continuous connectivity via satellite. The maritime and aviation industries have relied on satellite

communication for decades, and LEO constellations are now dramatically improving the speed and cost of these services (Scott, 2024).

- **Support for scientific and environmental monitoring:** Earth observation satellites collect continuous data on atmospheric conditions, ocean temperatures, ice coverage, deforestation rates, agricultural output, and dozens of other environmental variables. This monitoring capability is essential for climate science, disaster early warning systems, and sustainable resource management. No ground-based network could gather this data as efficiently or comprehensively (ESA, 2022).
- **Rapid scalability to new areas:** Extending a terrestrial network to a new area requires significant construction time and investment. Extending satellite coverage to a new area requires only deploying terminals, which can be done within days. This scalability makes satellites ideal for emergency response, seasonal or temporary coverage needs, and rapid expansion into newly developing markets (Congressional Budget Office, 2023).

5.2 Disadvantages of Satellite Communication

This suggests that while satellite communication has unique strengths, it also comes with a set of limitations that must be honestly acknowledged. Understanding these limitations is important for making informed decisions about when satellite communication is the right tool and when terrestrial alternatives are more appropriate (Roddy, 2019).

- **High capital and operational costs:** Designing, manufacturing, launching, and operating a satellite costs hundreds of millions to billions of dollars. Even relatively small LEO satellites, which are much cheaper than traditional GEO satellites, cost significant sums to build and launch in the quantities needed for a megaconstellation. These costs are ultimately passed on to consumers, making satellite services more expensive than comparable terrestrial services in areas where both are available (Roddy, 2019).
- **Signal delay for GEO systems:** The physical distance that GEO satellite signals must travel creates an unavoidable latency of approximately 500 to 700 milliseconds per round trip. For voice communication, this creates a noticeable echo effect that makes conversation feel unnatural. For interactive applications like online gaming, financial

trading, or video conferencing, the delay is a serious practical problem (Maral et al., 2020).

- **Vulnerability to rain fade:** Higher frequency signals used by modern satellite broadband systems can be significantly attenuated by heavy rain, thick clouds, and even heavy humidity. In tropical regions with frequent heavy rainfall, this can cause service interruptions at the times when weather events are most likely to affect people's lives. Engineers can mitigate rain fade through larger antennas, higher link margins, and adaptive modulation, but it cannot be entirely eliminated (Kratos Defense, 2023).
- **Limited bandwidth compared to fibre:** A single modern optical fibre cable can carry many terabits of data per second, which is orders of magnitude more than even the most advanced satellite system. In densely populated urban areas where fibre is available, satellite broadband cannot compete on raw bandwidth. Satellite's advantage lies in reaching places where fibre simply does not go (Viasat, 2022).
- **Fixed lifespan of satellites:** Satellites have a finite operational life, typically between 10 and 15 years for GEO satellites, determined largely by the amount of fuel they carry for station-keeping. When a satellite reaches the end of its life, it must be replaced, which requires planning years in advance and significant new investment. Unlike terrestrial infrastructure, which can be upgraded incrementally and repaired in place, a satellite in orbit cannot be serviced without extraordinary effort (Ilcev, 2017).
- **Contribution to orbital congestion and debris:** Every satellite launched adds to the population of objects in orbit. When satellites fail, collide, or are destroyed, they generate debris that can damage other satellites. The Kessler syndrome, a theoretical scenario in which a cascade of collisions generates so much debris that certain orbital regions become unusable, is considered a genuine long-term risk by space agencies. Managing this risk requires international cooperation and binding regulations that have proven difficult to establish (NASA, 2023).
- **Susceptibility to interference and jamming:** Satellite signals are broadcast through open space and can be intercepted, jammed, or spoofed by actors with appropriate equipment. Military adversaries have demonstrated the ability to jam satellite communications during conflicts. GPS spoofing incidents, where false navigation signals

mislead receivers, have been documented in both military and civilian contexts (Cisco, 2023).

5.3 Terrestrial Communication: Strengths and Limitations

When it comes to communication infrastructure, countries with advanced development often rely on systems like fibre optic networks, copper cable networks, and cellular mobile networks. These systems have some major advantages, especially when you look at how much data they can handle and how quickly they can send it. For example, a fibre optic link between two cities can carry a huge amount of data - hundreds of terabits per second - which is way more than any satellite system can manage. And the best part is that it can do all this with really low latency, meaning the delay between sending and receiving data is just a few milliseconds, even over short distances. This is a big deal, as it allows for fast and reliable communication, which is essential for many aspects of modern life.

In real life, it's usually cheaper to set up and run internet connections on the ground in cities because lots of people share the cost. For example, a special kind of cable that uses light to send data, called fibre optic, can be laid under a city street and provide internet to thousands of homes and businesses. This makes the cost per person really low. But in rural areas, it's a different story. The same length of cable might only serve a few people, making it too expensive for each person to pay for. This is where satellite internet comes in - it helps bridge this big gap in cost.

Satellites have a big advantage over regular networks on the ground - they're a lot less likely to get disrupted by physical problems. For example, floods can damage the cables that are underground, and big storms can knock down the towers that cell phones use. Earthquakes can even cut through the routes that fibre optic cables take, and fires can burn up the infrastructure that's above ground. But satellites don't have to worry about any of those things, because they're up in space. This is really important when it comes to planning for emergency communications, as NASA pointed out in 2023.

The way we communicate is changing, and it's not about choosing between satellites and traditional networks - it's about working together. A key part of planning for 6G is something called non-terrestrial networks, which combines satellite links with mobile networks. This means that instead of having a separate device for satellite connections, devices will be able to automatically switch between using ground-based networks and satellite links, depending on what's available and the strength of the signal. This approach is all about providing complete coverage everywhere, without having to build expensive infrastructure on the ground. It's a more flexible and efficient way to communicate, and it's being developed with the help of organizations like the NGMN Alliance, who are working on making this vision a reality by 2023. By combining the strengths of both satellites and traditional networks, we can create a more reliable and widespread communication system that benefits everyone.

6. Challenges Facing Satellite Communication

6.1 Atmospheric Attenuation and Signal Propagation

Atmospheric attenuation can be understood as the gradual weakening of an electromagnetic signal as it travels through the Earth's atmosphere. This happens because the atmosphere is not empty space. It is filled with gas molecules, water vapour, and suspended particles, all of which interact with radio waves in ways that absorb, scatter, or redirect their energy. The result is that a signal arriving at a satellite receiver is always somewhat weaker than the signal that left the transmitter, and the amount of weakening depends on the frequency of the signal and the current atmospheric conditions (Kratos Defense, 2023).

It can be observed that gaseous absorption is one of the primary mechanisms of attenuation. Oxygen molecules in the atmosphere absorb radio energy most strongly at frequencies around 60 GHz and 120 GHz, creating absorption peaks that satellite systems typically avoid. Water vapour absorbs strongly at around 22 GHz. Because modern satellite broadband systems use Ka-band frequencies in the 26.5 to 40 GHz range, they operate close to the water vapour absorption region, which means that humid atmospheric conditions and precipitation can significantly degrade signal quality (Kratos Defense, 2023).

Rain fade is the most practically significant form of atmospheric attenuation for commercial satellite systems. When rain is heavy, raindrops both absorb the energy of passing radio waves and scatter them in random directions away from the intended receiver. The severity of rain fade depends on the rainfall rate, the size of the rain drops, the elevation angle of the satellite link, and the frequency being used. Tropical regions, which experience intense convective rainfall, present a more challenging environment for Ka-band satellite services than temperate regions with lighter, more stratiform rain (Pratt et al., 2019).

In practice, satellite system designers address rain fade through a combination of approaches. Link margin, which means designing the system to transmit more power than the minimum required under clear sky conditions, provides a buffer that absorbs moderate levels of attenuation without service degradation. Adaptive coding and modulation automatically adjusts the efficiency of the signal encoding when conditions deteriorate, trading data rate for reliability to keep the link operational. Site diversity, where a user can switch to a backup ground station in a different weather zone, can effectively eliminate outages caused by local rain events for the most demanding applications (Viasat, 2022).

6.2 The Latency Challenge

When we talk about communication systems, there's something really important to consider: latency. This is basically the time it takes for data to travel from one point to another. For a lot of applications, latency is a big deal because even small delays can have a big impact. Imagine you're downloading a file - if it takes an extra second, you might not even notice. But if you're on a video call and there's a one-second delay, it can feel really weird and disrupt the whole conversation. And in some cases, like financial trading, a delay of just 100 milliseconds can be the difference between making a profit and missing out on an opportunity. It's interesting to think about how our perception of latency varies depending on what we're doing. Some delays are barely noticeable, while others can be really frustrating. This is something that researchers like Maral et al. have been studying, and their findings from 2020 highlight just how important it is to minimize latency in certain applications. In general, latency is a key performance characteristic of any communication system. It's not just about how fast data can travel, but also about how reliable and efficient the system is. By understanding and addressing latency, we can build better communication systems that meet our needs and help us stay connected.

GEO satellites have a big problem with latency, which is basically a delay in how long it takes for signals to travel between the satellite and the ground. This delay is because the satellite is about 35,786 kilometers away from the Earth, so even if signals are traveling at the speed of light, it still takes around 119 milliseconds for them to make a one-way trip. For a round trip, that's more like 240 milliseconds, and that's not even counting any extra delays that happen when the signal is being processed on the ground or routed through the internet. When you add all those extra delays together, the total latency for GEO satellite broadband is usually somewhere between 500 and 700 milliseconds per round trip. This has been a major limitation for using GEO satellites for interactive applications, like video calls or online gaming, because the delay can make it feel really slow and unresponsive.

Satellites in low Earth orbit, or LEO, have made a big difference in reducing latency for satellite broadband. When they're flying between 300 to 2,000 kilometers above the Earth, the time it takes for a signal to travel one way is just 1 to 7 milliseconds. This means that the total time for a round trip, or latency, can be as low as 20 to 50 milliseconds. That's similar to what you'd get with cable broadband, and it's good enough for things like video calls, online gaming, and most other interactive applications. But that's not all - satellites can also use laser links to talk to each other, which helps cut down on latency even more. This is especially useful for long-distance communications, because it lets data stay in space instead of having to go through multiple ground stations. According to a report by the Congressional Budget Office in 2023, this technology can really make a difference in how we communicate over long distances.

6.3 Orbital Debris and Space Sustainability

Orbital debris, sometimes called space junk, refers to all human-made objects in orbit that no longer serve any useful purpose. This includes defunct satellites whose batteries have died or whose fuel has been exhausted, spent rocket upper stages that were used to deliver payloads to orbit, and the fragments generated by collisions, explosions, and anti-satellite weapons tests. As of 2025, the U.S. Space Force's Space Surveillance Network was tracking nearly 50,000 objects in orbit, and the actual number of debris fragments too small to track but large enough to damage a satellite is estimated to be in the hundreds of thousands (NASA, 2023).

It can be observed that the orbital debris problem has worsened in recent years and is likely to worsen further as the number of satellites in orbit increases. In November 2021, Russia conducted a test of a direct-ascent anti-satellite missile, destroying its own defunct Cosmos 1408 satellite and creating approximately 1,500 trackable debris fragments and potentially hundreds of thousands of smaller pieces. This single event endangered the crew of the International Space Station and added significantly to the already crowded debris environment in low Earth orbit (NASA, 2023).

The theoretical scenario known as the Kessler syndrome describes a situation in which the density of debris in a particular orbital region becomes so high that every collision creates more debris, which causes more collisions, in a self-reinforcing cascade that eventually renders that orbital region unusable. Whether this threshold has already been crossed in some orbital bands is debated, but the concern is taken seriously by space agencies and satellite operators around the world (NASA, 2023).

This suggests that sustainable satellite operations require both technical and regulatory solutions. On the technical side, modern LEO satellites are typically designed to deorbit within five years of the end of their operational life, with Starlink satellites designed to do so passively through atmospheric drag at their operating altitudes. On the regulatory side, there is growing pressure from space agencies and international bodies for binding deorbit timelines, restrictions on intentional debris-generating events, and investment in active debris removal technologies (ITU, 2023).

6.4 Cybersecurity Threats

Satellites play a big role in the world's important systems, like money transfers, military commands, and emergency services. But as they become more important, they also become bigger targets for cyber attacks. The thing is, satellites are really hard to protect from these attacks. When a problem happens, you can't just fix it or replace the broken part like you can with systems on the ground. This is because satellites are in space, and it's really hard to get to them once they're up there. According to Cisco, this makes the cybersecurity challenges for satellite operators even tougher than for those who run systems on the ground.

One of the biggest dangers to satellites is when their ground control systems are compromised. You see, satellites are controlled from the ground by stations that send them commands and receive information through radio links. If someone with bad intentions can intercept and change those commands, or get into the computers at the ground station that send them, they could basically take over the satellite. Now, the links that send commands are encrypted, which means they're scrambled to keep them safe. But the problem is, some countries have gotten so good at cyber attacks that we can't just assume encryption is enough to protect us. For example, in 2022, a company called Viasat had its satellite internet service hacked right when Russia started invading Ukraine. This hack messed up communications for tens of thousands of people all over Europe, showing just how real and serious the threat of satellite cyber attacks is.

Satellite communications are not as secure as we think. Even if the messages are encrypted, it's still possible to find out who's talking to who, when, and how often. This can be a big problem. There's also something called GPS spoofing, where a fake signal is sent from the ground to override the real signal from the satellite. This can confuse ships and planes, and it's been happening in some sensitive areas around the world. The thing is, GPS was designed a long time ago, when this kind of spoofing wasn't a big concern. Now, it's a major weakness in our critical infrastructure, and that's a worry. According to Cisco, this is still a problem in 2023.

We can see that problems with the supply chain make things really tough to control. When you're building a satellite, you're getting parts from hundreds of different companies. If any of those parts are tampered with during manufacturing, testing, or when they're being put together, it's possible for someone to sneak in a bad actor before the satellite even gets into space. Also, the software that runs the satellites and the systems on the ground is incredibly complicated, and any weaknesses in that software can be used against us even after the system is up and running. And to make things worse, the new threat of quantum computers, which could eventually break through the codes we use now, adds another layer of worry when it comes to keeping our satellites safe from cyber threats.

7. Future Trends in Satellite Communication

7.1 Integration with 6G Mobile Networks

The next generation of mobile communications is what 6G is all about, it's like the next step after 5G. Right now, in 2024, the International Telecommunication Union is working on making 6G happen through its IMT-2030 framework, but we won't see it being used commercially until the 2030s. What's different about 6G is that it's not just about making our regular mobile networks faster and more reliable like 5G was, it's actually designed to connect everything, including satellites, into one big network that works everywhere. This means that no matter where you are, you'll have access to fast and reliable internet, which is a pretty big deal. The idea is to make it so that you can use your phone or device to connect to the internet from anywhere, whether you're in a city or out in the middle of nowhere, and it will all work seamlessly. It's an exciting development, and it could change the way we communicate and access information in a big way. Imagine a future where you're never really out of touch, no matter where you are. If you've got a device that can handle 6G, it'll just switch to a satellite connection if you're in an area without any cell towers. You won't even notice it's happening, and you won't have to lift a finger. The satellite link will be good enough for most things you want to do, maybe just a bit more expensive, until you get back to an area with regular coverage. The best part is, this whole process will happen automatically, without you even realizing it - the network will take care of everything.

From my understanding, the direct-to-device satellite capability that the FCC authorised in the United States in 2024 represents an early step in this direction. This authorisation allows LEO satellites to transmit directly to standard smartphones on certain frequency bands, initially limited to text messaging. Starlink and T-Mobile have already activated a version of this service for basic messaging in areas without cellular coverage. As the technology matures and devices are updated to take advantage of higher satellite bandwidths, this capability is expected to extend to voice and eventually high-speed data (Congressional Budget Office, 2023).

In practice, major technology companies and equipment manufacturers are making substantial investments in 6G research. Ericsson, Nokia, Huawei, Samsung, Apple, NTT Docomo, Bharti Airtel, and Reliance Jio have all announced active 6G research programmes. National governments are also investing through research funding agencies. The European Union, the United States, China, Japan, and South Korea each have funded national 6G initiatives. The outcome of these parallel efforts will shape the architecture of global communication networks for the 2030s and beyond (NGMN Alliance, 2023).

It can be observed that 6G is expected to deliver substantial improvements over 5G across several dimensions. Target peak data rates of 1 terabit per second are being discussed, compared to 5G's theoretical maximum of 20 gigabits per second. Air latency targets of 0.1 milliseconds would make 6G suitable for applications that require near-instantaneous response, such as remote surgery or truly autonomous vehicle coordination. Energy efficiency improvements would reduce the power consumption of both network infrastructure and connected devices. Reconfigurable intelligent surfaces, a technology that uses electronically controlled panels to redirect radio signals around obstacles, represent one of the more intriguing 6G research directions (NGMN Alliance, 2023).

7.2 Artificial Intelligence in Satellite Operations

Satellites are getting a big boost from artificial intelligence, and it's changing the game in ways that would have been hard to imagine just ten years ago. The main reason for this is scale - we're talking about constellations with 7,000 satellites, each making thousands of decisions every day. And when one wrong move can mean a collision that costs millions, it's clear that humans can't do it all on their own. AI isn't replacing human judgment, but it's taking care of the huge volume of routine decisions that need to be made quickly, freeing humans up to focus on the bigger picture. This is especially important when you consider that the consequences of a mistake can be catastrophic, and the speed at which decisions need to be made is incredibly fast. By handling the day-to-day decisions, AI is helping to make satellite operations more efficient and safer, which is a major step forward for the industry.

Collision avoidance is the most immediately critical application of AI in satellite operations. SpaceX reported that its Gen2 Starlink satellites performed approximately 84,990 propulsive manoeuvres during a six-month period between December 2024 and May 2025, with the majority executed to avoid other satellites or debris fragments. These manoeuvres are planned and executed autonomously, without waiting for human approval, because the timescales involved and the volume of potential conjunctions are too great for human operators to handle in real time. The AI systems that manage this process evaluate collision probabilities, assess available manoeuvre options, consider the impact on orbital positions and fuel reserves, and execute decisions within seconds (SpaceX, 2025).

Using artificial intelligence in space is becoming more popular, and one area that's growing fast is processing data onboard satellites. Usually, satellites collect raw data from sensors and send it back to Earth to be processed, which takes up a lot of bandwidth and causes delays between when something is observed and when we can actually use the information. But some companies, like Satellogic, are changing this by putting machine learning models right on their satellites. These models can look at images and pull out the important stuff before sending it back to Earth. This way, less data needs to be sent, and we get useful information faster. The satellite can even decide what data is worth sending, which is a big improvement. NASA has been working on this too, and it's an exciting development for the field.

We're seeing a new trend in the use of artificial intelligence - it's being used for something called predictive maintenance. Satellites are always sending back lots of data about how they're working, and this data can be used to figure out if something is going wrong. Usually, people look at this data and compare it to certain limits - if something goes over a limit, they get an alert. But with AI, we can look at the data in a more detailed way and find patterns that might mean something is about to go wrong. This can help us catch problems before they happen, which is really useful. For example, if we think a satellite is going to fail, we can make some changes to how it's working or even get a new satellite ready to replace it. This way, we can fix the problem before it causes any trouble, rather than waiting until after it's happened. NASA is already using this kind of technology, and it's proving to be really helpful.

This suggests that the long-term trajectory of satellite operations is toward increasing autonomy. The next generation of Starlink satellites, the V3 generation announced for launch at scale in 2026, is planned to include dedicated AI processing units capable of running complex inference models in orbit. The stated vision is to create a distributed orbital computing infrastructure, where satellites act not only as communication relays but as nodes in a globally distributed AI computing network. Whether this vision proves technically and economically viable at scale remains to be seen, but it points toward a future in which the distinction between a communication satellite and a computing platform becomes blurred (SpaceX, 2025).

7.3 Multi-Orbit Systems and the Digital Divide

In practice, the satellite industry is already looking beyond the LEO revolution toward multi-orbit architectures that combine GEO, MEO, and LEO satellites in integrated service offerings. The idea is to exploit the complementary strengths of each orbital regime: GEO's wide, stable coverage for broadcast and low-priority data; MEO's balance of coverage and latency for navigation-sensitive applications; and LEO's low latency and high speed for interactive broadband. By intelligently routing traffic across orbit types, operators hope to deliver services that are more reliable and flexible than any single orbit type could provide (Congressional Budget Office, 2023).

From my understanding, the most significant social opportunity created by the LEO satellite revolution is the potential to meaningfully reduce the global digital divide. According to estimates, more than 2.5 billion people worldwide lack reliable internet access, with the greatest concentrations in sub-Saharan Africa, South and Southeast Asia, and remote areas of Latin America. These are largely regions where terrestrial broadband has not been deployed because the economics do not work at the population densities and income levels involved. Satellite broadband, if it can be made affordable enough, offers a way to bypass the need to build terrestrial infrastructure altogether (ESA, 2022).

It can be observed that cost remains the primary barrier. At the time of writing, a Starlink residential kit costs several hundred dollars for the terminal hardware plus a monthly subscription fee of around $120 in most markets. While this is competitive with some terrestrial broadband options in developed countries, it is far out of reach for the majority of households in low-income countries. Achieving genuine affordability will require a combination of hardware cost reductions as manufacturing scales up, innovative financing or subsidy mechanisms, potentially including government or development bank funding, and creative business models such as shared community terminals that spread the cost across multiple users (Congressional Budget Office, 2023).

This suggests that the satellite industry's ability to close the digital divide will depend not only on technical progress but on decisions made by governments, international organisations, and the operators themselves about how to structure pricing and subsidy programmes. The analogy of the rural electrification programme in the United States during the 1930s, when government intervention was required to extend electricity to areas that the private sector would not serve profitably, is instructive. A similar public-private partnership model may be necessary to ensure that the benefits of LEO satellite broadband reach the communities that need it most rather than only those that can most afford it (Congressional Budget Office, 2023).

7.4 Space-Based Solar Power and Other Long-Term Possibilities

As we look to the future, satellites might be able to do a lot more than just help us communicate. One idea that's been around for a while is space-based solar power. This is where we collect solar energy in space, where the sun is always shining and there's no atmosphere to block the light. Then, we send that energy back to Earth as microwaves or laser beams. Space agencies have been studying this idea for decades. For example, Japan's space agency has done some experiments with sending power wirelessly. The European Space Agency has also looked into it, funding some conceptual studies. The idea itself is possible, but it's still really expensive to get all the equipment into space. This is because launch costs are so high right now.

As the cost of launching satellites continues to drop, it's likely that some projects that were once too expensive will become feasible. If the price per kilogram to reach orbit keeps falling, thanks to reusable rocket systems, we might see new opportunities emerge. This is similar to what happened with LEO broadband - it wasn't viable in 2000, but by 2020, it had become a reality. We can expect similar changes in other space-based services in the coming decades, maybe in the 2030s or 2040s. While it's hard to predict exactly what will happen, one thing is clear: cheaper access to space is a key factor in shaping the future of space technology, as noted by Goldman Sachs in their 2024 report. This trend will likely have a significant impact on the development of space-based services, making it possible for new companies and projects to enter the market. With the cost barrier lowering, we can expect to see more innovation and investment in space technology, leading to new applications and services that we can't yet imagine. As the industry continues to evolve, it will be exciting to see what new opportunities emerge and how they will shape the future of space exploration and development.

8. Conclusion

From my understanding, satellite communication has completed a remarkable journey from its origins in the space race of the 1950s and 1960s to its current status as essential global infrastructure. What began as an experimental technology developed by superpowers competing for geopolitical prestige has evolved into a commercial industry serving billions of people in applications ranging from television broadcasting to precision agriculture to emergency disaster response. The story of satellite communication is, in many ways, the story of how a capability developed for one purpose is again and again repurposed and reimagined as technology improves and costs fall.

It can be observed that the current moment in satellite history is particularly significant. The shift from a handful of large, expensive geostationary satellites to constellations of thousands of smaller, cheaper LEO satellites is not merely an incremental improvement but a fundamental architectural change. This change is driving down the cost of connectivity for remote and rural users, improving the performance of satellite broadband to levels competitive with terrestrial alternatives, and opening up new applications in IoT connectivity, direct-to-device mobile service, and onboard satellite computing that were not practical with previous generations of technology.

This suggests, however, that significant challenges remain. The rapid proliferation of satellites is straining the regulatory frameworks that govern spectrum allocation and orbital slot management. The growing population of objects in orbit is increasing collision risks and the long-term sustainability of satellite operations. Cybersecurity vulnerabilities in satellite systems represent genuine risks to critical infrastructure. And the economic barriers that prevent the world's poorest and most remote communities from benefiting from satellite broadband have not yet been adequately addressed by either the private sector or governments.

In practice, addressing these challenges will require coordinated action on multiple fronts simultaneously. Engineers must continue developing more capable and more affordable satellite hardware. Regulators must adapt existing frameworks, many of which were designed for a world of a few dozen geostationary satellites, to handle the reality of tens of thousands of LEO satellites. International bodies must establish and enforce standards for responsible behaviour in orbit. Governments and development institutions must find ways to make satellite connectivity affordable for the communities that stand to benefit most from it.

Looking ahead, the integration of satellite networks with 6G mobile systems, the application of artificial intelligence to satellite operations, and the development of multi-orbit hybrid architectures all point toward a future where reliable, high-quality connectivity is available to essentially everyone on Earth, regardless of where they happen to live. Whether that future arrives quickly or slowly, and whether it reaches everyone or only those who can afford it, will depend on the decisions made by technologists, policymakers, and business leaders in the years immediately ahead. The technical foundations for universal connectivity are being laid right now. What is built on those foundations is still to be determined.

From my understanding, satellite communication will remain one of the most dynamic and consequential fields in telecommunications for the foreseeable future. It is a field where the boundaries of what is technically possible continue to move rapidly, where the economic dynamics are shifting in ways that create new opportunities, and where the social stakes, particularly around universal access to information and communication, could not be higher. It is a field worth following closely (Maral et al., 2020; Roddy, 2019; Congressional Budget Office, 2023).

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